

ArbitrageAI Platform — Non-Functional Requirements Document (NFRD)

Classification: Internal — Engineering
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Prepared for: Technical Lead & Engineering Team

1. Performance Requirements

1.1 Latency

ID	Requirement	Target	Measurement
PERF-01	Trade signal-to-order latency	p99 < 500ms	Cloud Monitoring: signal_emit_ts → order_submit_ts
PERF-02	Spread detection latency	< 50ms from WebSocket price update to signal	Cloud Trace
PERF-03	Vertex AI inference latency	< 50ms per prediction	Vertex AI prediction latency metric
PERF-04	Dashboard page load	p95 < 2s web; p95 < 1.5s mobile	Web Vitals (LCP); Firebase Performance SDK
PERF-05	P&L update to dashboard	< 3s from trade close to refresh	Firestore listener propagation time
PERF-06	Exchange connection validation	< 2s from API key paste to confirmation	Backend API response time
PERF-07	WhatsApp notification delivery	< 5s from trade close to message received	Cloud Run dispatcher latency
PERF-08	SMS delivery	< 15s from trigger to received	Twilio delivery timestamp
PERF-09	Subscription activation	< 60s from payment to tier active	Webhook → Firestore update latency
PERF-10	Exchange reconnection	< 30s after WebSocket disconnect	CCXT Pro reconnect timing

1.2 Throughput

ID	Requirement	Target
THRU-01	Concurrent active strategy workers	1,000,000
THRU-02	Strategy evaluations per second	500,000

ID	Requirement	Target
THRU-03	Trade records written per day	50,000,000
THRU-04	Exchange API calls per minute	200,000
THRU-05	Concurrent dashboard WebSocket sessions	1,000,000
THRU-06	WhatsApp notifications per second (burst)	10,000
THRU-07	Firestore reads per second	1,000,000

2. Availability & Reliability

2.1 Uptime SLOs

ID	Service	Target SLO	Window
AVAIL-01	Control plane — Starter/Pro	99.9%	Rolling 30 days
AVAIL-02	Control plane — Institutional/Enterprise	99.95%	Rolling 30 days
AVAIL-03	Strategy execution workers	99.9%	Rolling 30 days
AVAIL-04	Exchange Connector Service	99.9%	Rolling 30 days
AVAIL-05	Notification dispatcher	99.5%	Rolling 30 days
AVAIL-06	BigQuery analytics pipeline	99%	Rolling 30 days

2.2 Recovery Objectives

ID	Metric	Target
RECO-01	RTO — P1 incidents	< 15 minutes
RECO-02	RPO — trade data	< 1 minute (Cloud Spanner synchronous)
RECO-03	RPO — user config	< 5 minutes (Firestore multi-region)
RECO-04	Strategy worker pod restart	< 30s
RECO-05	KILL_SWITCH to all workers paused	< 10s

2.3 Fault Tolerance

ID	Requirement	Implementation
FAULT-01	No SPOF in control plane	GKE Autopilot multi-zone; min 3 replicas
FAULT-02	Exchange connector failover	REST fallback within 5s of WebSocket failure
FAULT-03	Vertex AI fallback	Static allocation model if online prediction unavailable
FAULT-04	Notification channel fallback	WhatsApp failure → SMS within 30s
FAULT-05	FX rate failover	Wise API fallback within 5s

ID	Requirement	Implementation
FAULT-06	Database write failure	Cloud Spanner write retried 3x with exponential backoff

3. Scalability

ID	Requirement	Target	Implementation
SCALE-01	Horizontal pod autoscaling	0 to 1M concurrent workers	GKE Autopilot HPA on CPU + Pub/Sub queue depth
SCALE-02	Database scaling	50M trade writes/day	Cloud Spanner linear horizontal scaling
SCALE-03	User registration	10M registered users	Firebase Auth (100M+ capacity)
SCALE-04	Zero-cost idle users	Near-\$0 for inactive users	GKE Autopilot scale-to-zero
SCALE-05	Africa traffic spikes	10x spike during naira devaluation events	GKE Autopilot auto-provision; Cloud CDN edge cache
SCALE-06	BigQuery analytics	Sub-10s P&L queries	Partitioned by user_id + date; clustered by strategy_type

4. Security

4.1 Data Security

ID	Requirement	Implementation
SEC-01	Credentials encrypted at rest	GCP Secret Manager AES-256
SEC-02	Credentials encrypted in transit	TLS 1.3 enforced by Cloud Load Balancer
SEC-03	Credentials never in application database	Secret Manager only; Firestore stores reference paths only
SEC-04	Credentials never logged	Log sanitization; structured log schema excludes credential fields
SEC-05	Credentials never returned to frontend	API responses never include raw API keys
SEC-06	User data encrypted at rest	Google-managed encryption (Firestore + Cloud Spanner)
SEC-07	PII isolation	User PII separate from trade data in different Firestore collections

4.2 Access Control

ID	Requirement	Implementation
SEC-08	Workload Identity	All pods use Workload Identity; zero static service account keys
SEC-09	Namespace isolation	GKE NetworkPolicy prevents cross-namespace communication
SEC-10	Secret Manager scoping	Each namespace SA can only read its own /users/{uid}/ secrets

ID	Requirement	Implementation
SEC-11	API authentication	Firebase Auth JWT on every API endpoint
SEC-12	Admin access	Separate admin role claim in Firebase Auth custom tokens
SEC-13	VPC-SC perimeter (Institutional+)	Data plane cluster in VPC Service Controls perimeter

4.3 Application Security

ID	Requirement	Implementation
SEC-14	Input validation	Zod schema validation (TypeScript); server-side on all APIs
SEC-15	WAF protection	Cloud Armor; OWASP Top 10 ruleset
SEC-16	Rate limiting	Per-IP: Cloud Armor; per-user: API middleware
SEC-17	SSRF protection	Exchange Connector only connects to whitelisted domains
SEC-18	Dependency scanning	GitHub Dependabot + Snyk in CI/CD; no deploy with critical CVEs
SEC-19	Secrets scanning	GitHub secret scanning; pre-commit hook blocks credential commits
SEC-20	Penetration testing	Annual third-party pentest; required before Institutional launch

5. Compliance

ID	Requirement	Jurisdiction	Timeline
COMP-01	FSCA CASP registration	South Africa	Before ZA public launch
COMP-02	SEC Nigeria Digital Asset registration	Nigeria	Before NG public launch
COMP-03	Kenya VASP Act registration	Kenya	Before KE public launch
COMP-04	GDPR compliance	EU	Before EU user acceptance
COMP-05	POPIA compliance	South Africa	Before ZA public launch
COMP-06	NDPA compliance	Nigeria	Before NG public launch
COMP-07	FinCEN MSB legal opinion	USA	Before US marketing launch
COMP-08	SOC 2 Type II	Global (Institutional)	Month 12
COMP-09	SARS-compliant tax export	South Africa	Phase 3
COMP-10	Data retention policy	Global	12-month active; 7-year archived
COMP-11	Non-custodial legal opinion	Global	Before any launch
COMP-12	Nigeria P2P fiat leg legal review	Nigeria	Before automating fiat leg (Phase 5)

6. Observability

ID	Requirement	Implementation
OBS-01	Distributed tracing	Cloud Trace via OpenTelemetry SDK
OBS-02	Structured logging	JSON logs to Cloud Logging from all services
OBS-03	Custom metrics	Trade execution latency, signal latency, execution quality score
OBS-04	SLO dashboards	Cloud Monitoring SLO tracking
OBS-05	Error tracking	Sentry (frontend + backend)
OBS-06	Product analytics	PostHog (onboarding funnel, retention cohorts)
OBS-07	Alerting	PagerDuty: P1/P2 (24/7 on-call); Slack: P3/P4
OBS-08	Audit logging	All admin actions + secret access attempts; immutable

7. Maintainability

ID	Requirement	Implementation
MAINT-01	Infrastructure as Code	All GCP resources in Terraform; no manual provisioning
MAINT-02	Kubernetes workload management	All GKE workloads in Helm charts
MAINT-03	CI/CD pipeline	GitHub Actions → Cloud Build → GKE rolling deploy
MAINT-04	Zero-downtime deploys	GKE rolling updates with health checks + PodDisruptionBudget
MAINT-05	Feature flags	LaunchDarkly or GCP Remote Config
MAINT-06	Database migrations	Flyway for Cloud Spanner DDL; backward-compatible only
MAINT-07	Code quality	ESLint + Prettier (TypeScript); Ruff (Python); enforced in CI
MAINT-08	Test coverage	Min 80% unit test coverage; strategy logic requires 95%
MAINT-09	Documentation	All APIs in OpenAPI 3.0; all services have README + runbook

8. Usability & Accessibility

ID	Requirement	Target
UX-01	Onboarding completion rate	> 85% of signups activate AI engine
UX-02	Exchange connection time	< 4 minutes wizard start to validated connection
UX-03	Time to first trade	< 24 hours from AI engine activation
UX-04	Mobile-first design	All screens functional at 375px; tested on Android Go
UX-05	Low-bandwidth mode	Core dashboard on 3G (< 1MB page weight)
UX-06	Offline state	Cached P&L shown with "last updated" indicator
UX-07	Accessibility	WCAG 2.1 AA; VoiceOver/TalkBack support

ID	Requirement	Target
UX-08	Localization	English launch; French Phase 2; Portuguese Phase 3
UX-09	Zero jargon rule	No technical trading terms without plain-language explanation

9. Africa-Specific Non-Functional Requirements

ID	Requirement	Target
AFR-01	GCP region for South Africa	africa-south1 (Johannesburg); < 50ms to VALR/Luno
AFR-02	Nigeria/West Africa routing	europa-west1 until GCP Lagos available
AFR-03	WhatsApp primary notification	100% of NG/KE/GH/TZ/UG/RW users; SMS fallback within 30s
AFR-04	Mobile money payment success	> 95% successful confirmations
AFR-05	NGN exchange rate freshness	P2P rate updated every 5 minutes
AFR-06	Low-end Android performance	Usable on Android 9+ with 2GB RAM
AFR-07	Africa P2P data availability	Historical data collector starts Day 1 of build
AFR-08	SARS tax export	Available by Phase 3

10. Disaster Recovery

ID	Scenario	Response Plan	RTO	RPO
DR-01	africa-south1 outage	Failover to europe-west4	< 5 min	< 1 min
DR-02	Cloud Spanner regional outage	Multi-region auto-failover	< 1 min	0
DR-03	Vertex AI endpoint down	Static allocation model activates	< 1 min	N/A
DR-04	Exchange API outage (Binance)	Binance strategies paused; others continue	< 5 min	N/A
DR-05	Stripe outage	Flutterwave backup; new subs queued	< 30 min	N/A
DR-06	Complete platform outage	KILL_SWITCH; status page; SMS all users	< 15 min	< 1 min
DR-07	Security breach	Affected secrets rotated; users notified; legal engaged	< 30 min	N/A

11. Cost Management

ID	Requirement	Implementation
COST-01	GKE scale-to-zero for inactive users	Worker pods terminated when engine inactive
COST-02	BigQuery cost control	Query cost estimates enforced before dashboard queries
COST-03	Vertex AI cost monitoring	Budget alerts at 80% of monthly budget
COST-04	Notification cost monitoring	Alert if WhatsApp cost/user > \$0.05/day

ID	Requirement	Implementation
COST-05	GCP budget alerts	Alerts at 50%, 80%, 100% of monthly budget
COST-06	Cost per user tracking	Monthly cost tracked per tier; alert if margin < 70%

NFRD v1.0 — June 4, 2026. All requirements subject to review by Tech Lead prior to Phase 1 kickoff.